

Discussion

* How many positives reviews are misclassified as negatives?

→ I got "863" Positive reviews misclassified as negative and "902" negative misclassified as positive reviews – Logistic Regressions (LR)

→ I got "807" Positive reviews misclassified as negative and "738" negative misclassified as positive reviews – Naïve Bayes (NB)

* Which type of Error is more harmful?

→ In this task we have two types of errors, false negatives and false positives. A false negative happens when reviews is actually positive, but model predicts it negative, which means the system fails to recommend a good movie. As a result, the user may never discover a movie they would really enjoy. However, from a business point of view, this can reduce user engagement and possibly lead to loss of revenue. Therefore, it has high impact.

On the other hand, false negative happens when the review is actually negative, but model says it positive. In this case, the system recommends a bad movie and the user might waste time watching something they do not like, and this could reduce the trust in the system. Therefore, this impact is usually moderate.

Class Room Exercise

→ Top 10 Most Positives words:

- great, love, best, excellent, wonderful well, enjoy, beautiful, perfect, lov - LR
- custor, mattau, prot, felix, bourne, perfection, flawless, Miyazaki, lemon, understand – NB

→ Top 10 Most Negative words:

- bad, not, worst, no, movie, waste, awful. poor, even, bor - LR
- uwe, boll, stinker, waste, mst, gadget, unwatchable, unfunny, redeeming, Norris – NB

* Show Misclassified Examples

Review 05: I've seen a fair few times from the Far East recently..... Some were Excellent (Battle Royale, Infernal affairs, the Eye), and some were not great.....

Top Influencing words: LR

- great (+), not (-), bar (-), excellent (+), could (-)
- Actual Positive: Predicted Negative

Review 10: I give these 3 stars out of a possible 10 because the stories are open-ended and left unexplained, and because of the nauseating scenes of someone eating in an extremely disgusting way.....

Top Influencing words: NB

- Disgusting (-), dylan (+), helen (+), lonely (+), none (-)
- Actual Negative: Predicted Positive

* **Negation:** Yes, negation fools the model. For Example, when a review says "not good" the model sees "good" as positive word and ignores the "not" that completely revere the meaning. This is because the BoW models treat words as independent items and cannot understand that "not" changes the words that that come after it.